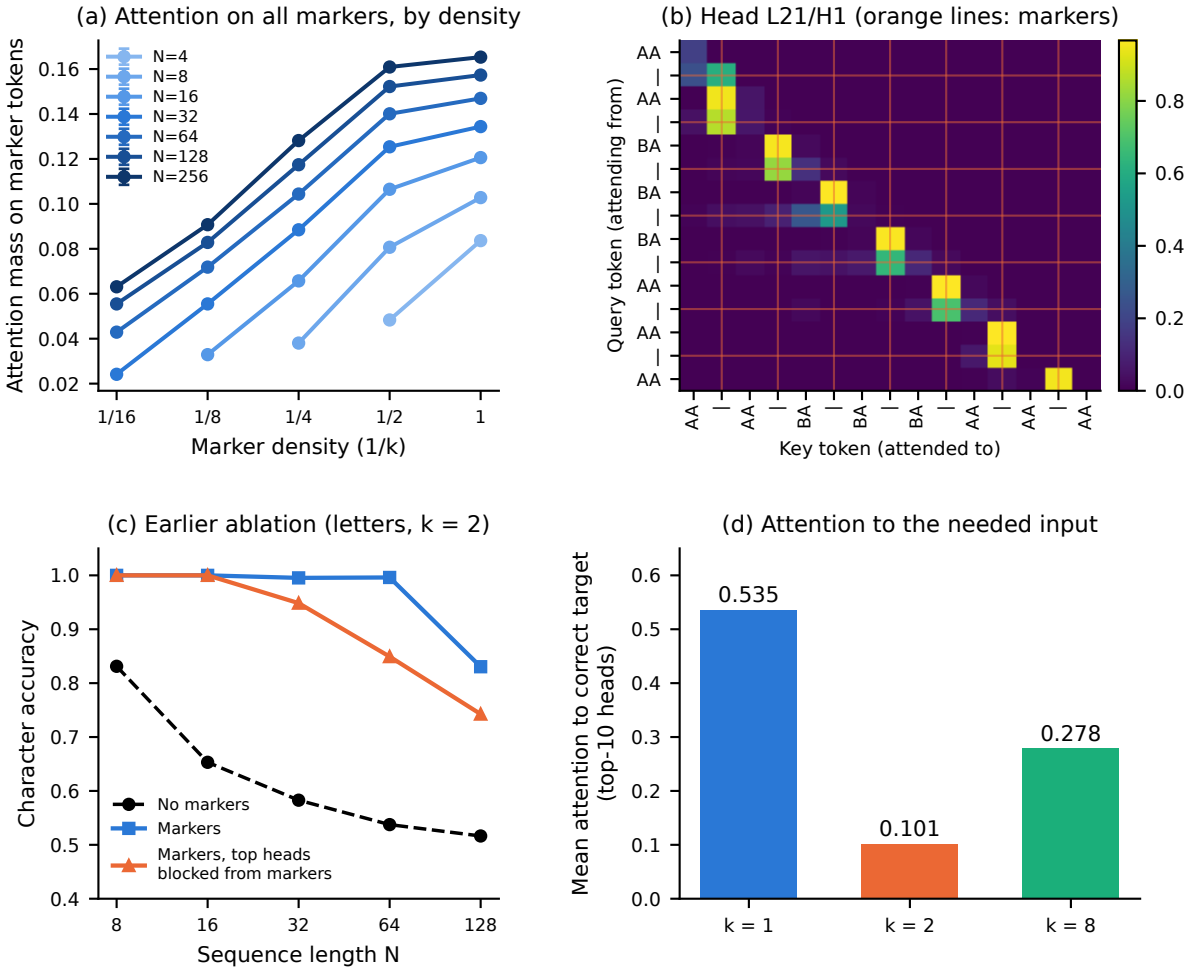
(a) Attention on all markers, by density
Attention mass on marker tokens
Marker density (1/k)
N=4
N=8
N=16
N=32
N=64
N=128
N=256
1/16
1/8
1/4
1/2
1
(b) Head L21/H1 (orange lines: markers)
Query token (attending from)
Key token (attended to)
(c) Earlier ablation (letters, k = 2)
Character accuracy
Sequence length N
No markers
Markers
Markers, top heads blocked from markers
(d) Attention to the needed input
Mean attention to correct target (top-10 heads)
0.535
0.101
0.278
k = 1
k = 2
k = 8